TIC 59453672

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

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BLS EPHEMERIS
  Period       4.178961 d  (ALIASED, see below)
    -> corrected period: 8.357922 d
  Epoch (t0)   2503.3878 BTJD
  Duration     0.1200 d
  Depth        2.149 %  (21,488 ppm)
  SNR          239.0

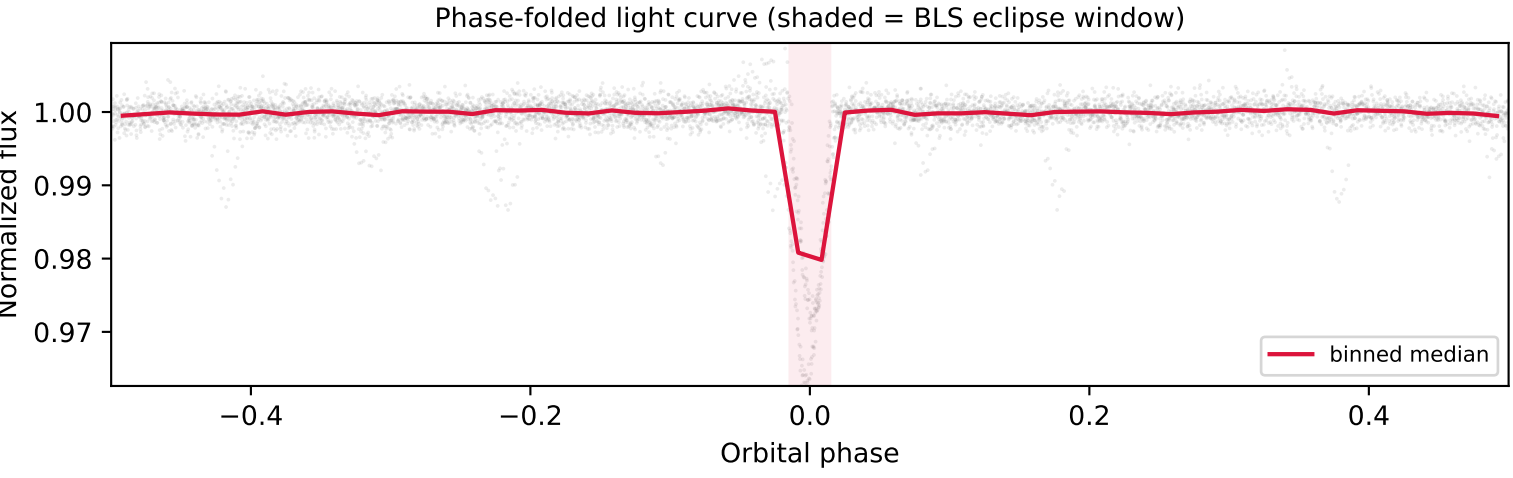
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     12  (gate >= 3)
  Robust depth ratio    1.06  (gate >= 0.50)
  Robust depth sigma    185.0  (gate >= 7.0)
  Odd/even z-score      23.67  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

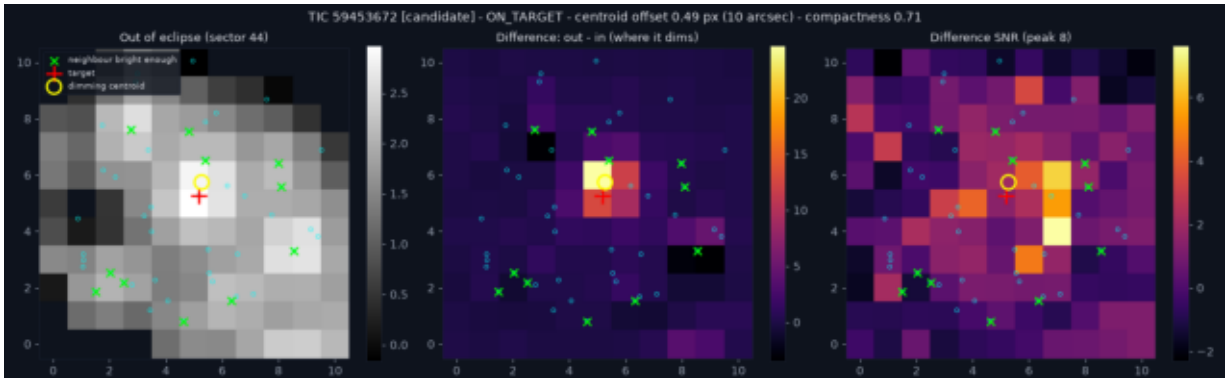
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 44    light-curve dip in this sector: 116.3 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   7.63   (gate >= 5.0)      Compactness   0.71   (gate >= 0.50)
  Dimming centroid offset     0.49 px (10 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 59363880
    0.80 px away, dTmag +3.55 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 23.67 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 8.357922 d, in any write-up.